

Multimodal AI Services

> Relevant source files

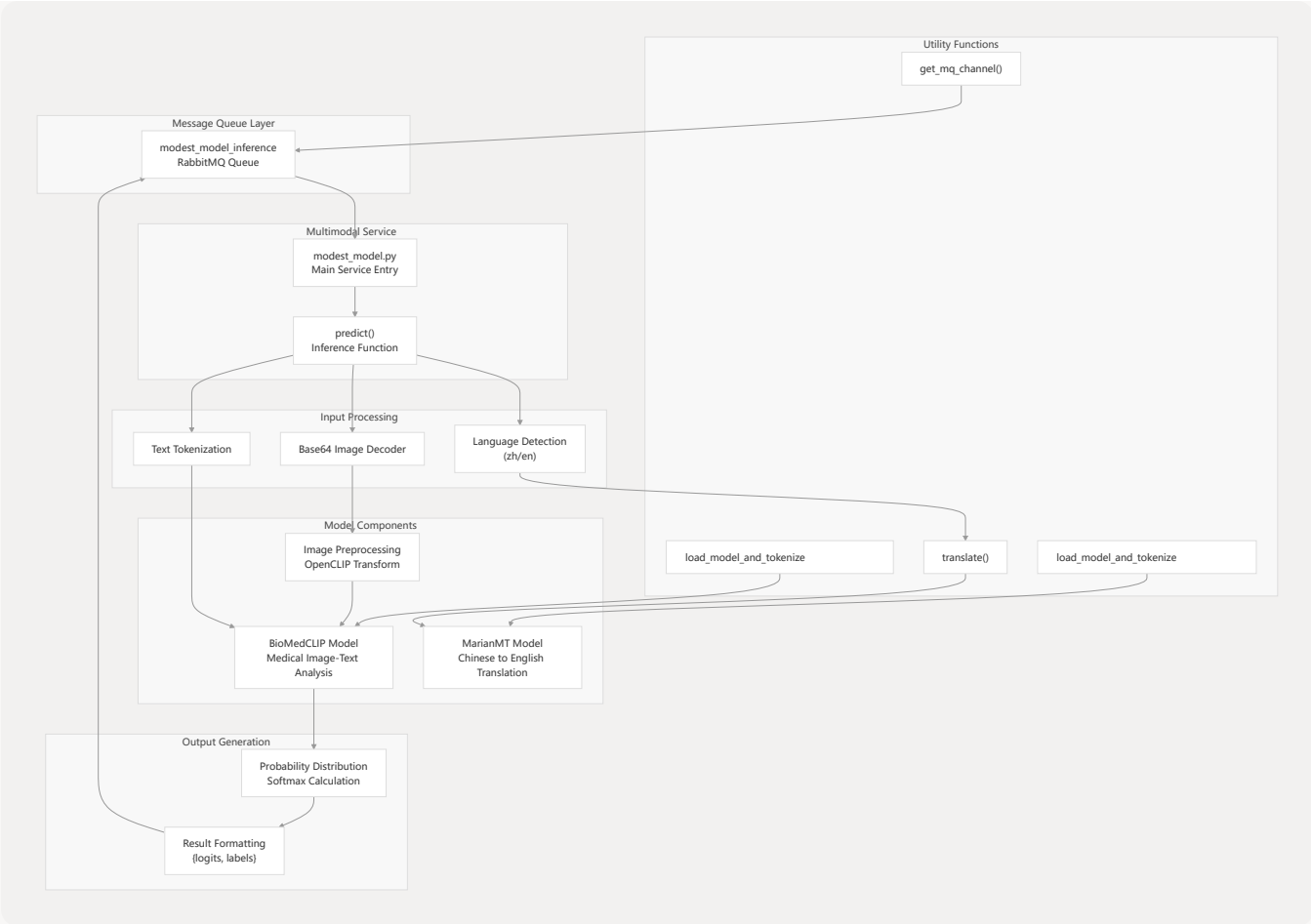
This page documents the multimodal AI services that provide medical image and text analysis capabilities in the MKTY Smart Healthcare System. These services integrate BioMedCLIP for medical image-text analysis and MarianMT for translation, enabling multilingual multimodal medical diagnosis.

For large language model services, see [Large Language Model Service](#). For time series and text analysis services, see [Time Series & Text Analysis](#).

Purpose and Scope

The multimodal AI services handle the processing of medical images combined with textual descriptions to provide diagnostic insights. The system supports both Chinese and English inputs, automatically translating Chinese text to English when needed for optimal model performance. The service operates as a message queue consumer, processing requests asynchronously via RabbitMQ.

Architecture Overview



Sources: `backend/modest_model.py` 1-81 `backend/modest_model_util.py` 1-184

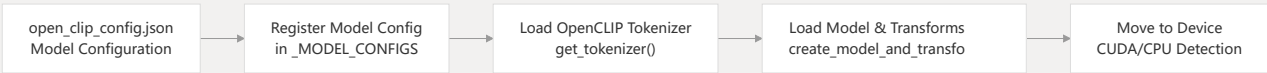
Core Model Components

BioMedCLIP Model

The BioMedCLIP model provides medical image-text analysis capabilities using the OpenCLIP framework. This model is specifically designed for biomedical applications and can analyze medical images in relation to textual descriptions.

Component	Details
Model Name	BiomedCLIP-PubMedBERT_256-vit_base_patch16_224
Framework	OpenCLIP (not yet fully supported by transformers)
Input	Medical images + text labels
Output	Probability distributions across text labels
Device Support	CUDA/CPU automatic detection

Model Loading Process:



The model loading is handled by the `load_model_and_tokenizer_BiomedCLIP` function which:

1. Loads configuration from `open_clip_config.json`
2. Registers the model configuration if not already present
3. Creates tokenizer using `get_tokenizer`
4. Loads the model using `create_model_and_transforms` with pretrained weights
5. Sets up image preprocessing pipeline
6. Moves model to appropriate device (GPU if available, otherwise CPU)

Sources: `backend/modest_model_util.py` 102-146

MarianMT Translation Service

The MarianMT model provides Chinese to English translation to ensure optimal performance with the BioMedCLIP model, which is primarily trained on English medical texts.

Component	Details
Model	<code>Helsinki-NLP/opus-mt-zh-en</code>
Framework	Transformers (Hugging Face)
Purpose	Chinese text → English text translation
Auto-download	Yes, via Hugging Face Hub

Translation Process Flow:

Chinese Text Labels
List[str]



Language Detection
if language == 'zh'

Chinese



Tokenize Text
tokenizer([text])

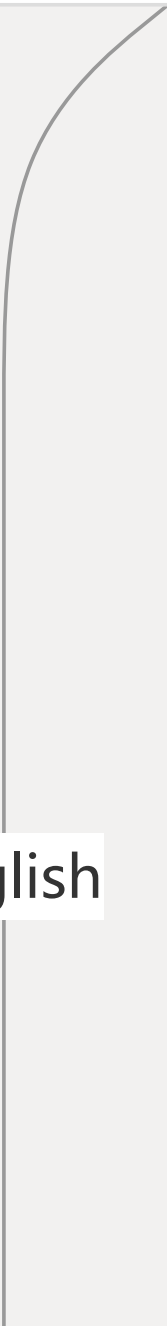


Model Generation
model.generate(**batch)



Decode Translation

English



batch_decode()

English Text Labels
List[str]

The translation is performed by the `translate` function which processes text in batches and handles multiple text inputs simultaneously.

Sources: `backend/modest_model_util.py` 148-183 `backend/modest_model.py` 54-55

Message Queue Integration

The multimodal service operates as a RabbitMQ consumer, processing inference requests asynchronously. This design enables scalable processing and decouples the AI inference from the main web application.

Queue Configuration

Parameter	Value	Purpose
Queue Name	<code>modest_model_inference</code>	Identifies the multimodal service queue
Host	<code>localhost</code>	RabbitMQ server location
Port	<code>5672</code>	Standard RabbitMQ port
Durable	<code>True</code>	Queue persists across server restarts
Prefetch Count	<code>1</code>	Process one message at a time

Message Format

Input Message Structure:

```
{  
  "texts": ["label1", "label2", "label3"],  
}
```

```
"image_base64": "data:image/jpeg;base64,/9j/4AAQ...",  
"language": "zh"  
}
```

Output Message Structure:

```
{  
  "logits": [0.1, 0.7, 0.2],  
  "labels": ["translated_label1", "translated_label2", "translated_label3"]  
}
```

Sources: `backend/modest_model.py` 20-34 `backend/modest_model_util.py` 71-100

Processing Pipeline

The complete multimodal inference pipeline processes both image and text inputs to generate probability distributions across diagnostic labels.

